

Quiz 2

① $R=2$ $\sigma_s^2 = 0.25$ $\sigma_s = \sqrt{0.25}$

$$49 \text{ dB} = 6.02N + 10.8 - 20 \log\left(\frac{2}{\sqrt{0.25}}\right) \Rightarrow N = 8.34$$

$$\Rightarrow \boxed{N=9} \text{ bits}$$

② $y(t) = \cos(100\pi t)$ I don't know what "p" means, I assume 100π

$$\hookrightarrow y(t) = \cos(100\pi t)$$

sampled @ 90 Hz

$$f_0 = 50 \text{ Hz}$$

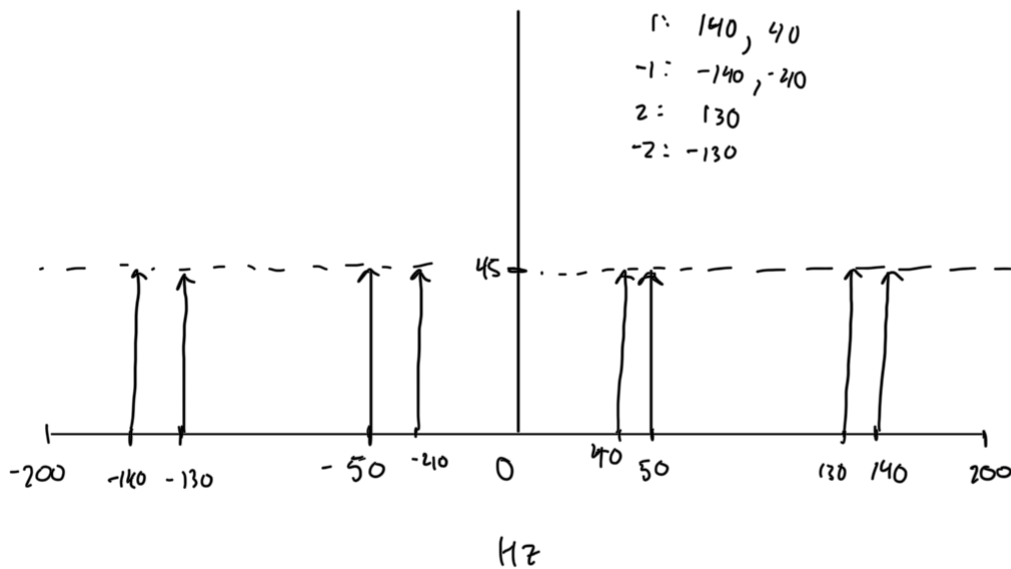
$$\pm 50 + k(90)$$

$$1: 140, 40$$

$$-1: -140, -40$$

$$2: 130$$

$$-2: -130$$



$$\underline{\pm 50, \pm 40, \pm 130, \pm 140}$$